

Abby R. Mirsky

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EDUCATION

Duke University Pratt School of Engineering | Durham, NC

B.S. in Mechanical Engineering | GPA: 3.73/4.0

- **Relevant Courses:** Mechatronics, Dynamics, Mechanics of Solids, Engineering Innovation (Mechanical Engineering), Special Topics in Engineering, Engineering Design and Technical Communication

TECHNICAL SKILLS | **CAD:** SolidWorks, Autodesk Inventor, Onshape • **Analysis:** FEA (stress analysis), Design Validation, Testing & Evaluation, Excel • **Programming:** Python, C++, Arduino (embedded systems) • **Tools:** Git, GIS, Word, PowerPoint.

PROJECTS

Underwater Robotics Design Team | *Engineering Project*

Aug. 2025 - Present

- Lead **mechanical design** and **subsystem integration** of submersible **robotic platform (8 ft depth)**, coordinating hull structure, propulsion, sealing, and retrieval mechanisms.
- Integrated motors and **Arduino-based** control architecture to ensure synchronized actuation and system reliability.
- Conduct **iterative testing** and failure analysis to improve structural integrity and operational stability.
- Developed mechanical design criteria and evaluated subsystem feasibility through **iterative testing**.

Solid Propulsion Design | *Duke AERO*

Sept. 2025 - Present

- Performed structural **FEA** on high-pressure **motor components** to validate safety factor (**FoS = 3**) under **hydrostatic** and **static-fire** load cases.
- Redesigned sealing interfaces and integrated pressure sensors for real-time system monitoring.
- Supported test preparation, design reviews, and safety evaluations aligned with engineering standards.

Liquid Medication Dispenser | *Engineering Project*

Aug. 2024 - Dec. 2024

- Worked with a **client** at Duke Hospital to design an integrated mechanical dispensing system with **Arduino-controlled** actuation for reliable timed fluid delivery for patients requiring consistent medication adherence.
- Coordinated **mechanical-electrical integration** to ensure controlled dosage accuracy and **user safety**.
- Prototyped using digital fabrication; refined design through iterative performance testing.
- Prepared **technical reports** and **presentation materials** for internal design reviews.

WORK EXPERIENCE

The Ryan Lab | Durham, NC | *Undergraduate Research Assistant*

Jan. 2026- Present

- Supporting **mechanical design** of a field-deployable robotic system to quantify glacial meltwater runoff in Greenland for **climate research**.
- Contribute to structural design and hardware development for operation in high-stress glacial stream environments.
- Assist with laboratory testing preparation and iterative mechanical refinement to improve **system reliability**.

Engineers In Action | Shiselweni Region of Eswatini, Africa | *Intern*

Dec. 2024 - July 2025

- Collaborated on construction of **128m** pedestrian **suspension footbridge** supporting **long-term public** use.
- Coordinated **field construction** activities including **cable sag** setup, **masonry**, and **deck installation**.
- Developed and adapted **safety protocols** to maintain compliance with engineering safety standards during construction.
- Performed field verification, quality control, and real-time issue resolution.

HRS Water Consultants | Lakewood, CO | *Intern*

June. 2023- Aug. 2023

- Supported groundwater research and pollution analysis by assisting with **fieldwork**, logging and processing environmental data, and supporting development of **GIS maps** to visualize water quality and conservation trends.
- Digitized the firm's physical engineering records into a searchable **digital library** to improve data accessibility.

OTHER EXTRACURRICULARS & AFFILIATIONS

US Figure Skating Association (USFSA) Solo Dance **National champion**/6-time competitor • DejaBlue Accapela music **composer** • DukeAERO • Society of Women Engineering (SWE).